

In-Class Assignment 01

Student Number – 35040

Q1

```
public class car {
    int year;
    String brand;
    double price;

    car() {
        year = 2000;
        brand = "Unknown";
        price = 0.00;
    }

    void displayInfo() {
        System.out.println(x: "Car Information ");
        System.out.println("Car Year :" + year);
        System.out.println("Car Brand:" + brand);
        System.out.println("Car Price " + price);
    }

    Run | Debug
    public static void main(String[] args) {
        car car1 = new car();
        car1.displayInfo();
    }
}
```

```
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Car Information  
Car Year :2000  
Car Brand:Unknown  
Car Price 0.0
```

Q2

```
public class Student {  
    String StudentID;  
    String name;  
    int age;  
    double gpa;  
  
    public Student(String studentId, String name, double gpa, int age) {  
        this.StudentID = studentId;  
        this.name = name;  
        this.gpa = gpa;  
        this.age = age;  
    }  
  
    void introduce() {  
        System.out.println("Hello, my name is " + name + ". I am " + age + " years old and my GPA is " + gpa + ".");  
    }  
  
    void updateGpa(double newGpa) {  
        gpa = newGpa;  
        System.out.println("GPA updated to: " + gpa);  
    }  
  
    Run | Debug  
    public static void main(String[] args) {  
        Student s1 = new Student(studentId: "S12345", name: "Navitha", gpa: 3.5, age: 19);  
        Student s2 = new Student(studentId: "S67890", name: "Akitha", gpa: 3.2, age: 22);  
  
        s1.introduce();  
        s1.updateGpa(newGpa: 3.8);  
  
        s2.introduce();  
        s2.updateGpa(newGpa: 4);  
    }  
}
```

Q3

```
public class product {
    String productID;
    String productName;
    double unitPrice;

    product() {
        productID = "0";
        productName = "Unknown";
        unitPrice = 0.00;
    }

    product(String id, String name, double price) {
        productID = id;
        productName = name;
        unitPrice = price;
    }

    void applyDiscount(double percentage) {
        unitPrice = unitPrice - (unitPrice * percentage / 100);
        System.out.println("Updated Price after " + percentage + "% discount: Rs." + unitPrice);
    }

    String getProductLabel() {
        return productID + " - " + productName + " - Rs." + unitPrice;
    }
}

Run | Debug
public static void main(String[] args) {
    product p1 = new product();

    product p2 = new product(id: "101", name: "Laptop", price: 150000);

    p2.applyDiscount(percentage: 15);

    System.out.println("Product 1: " + p1.getProductLabel());
    System.out.println("Product 2: " + p2.getProductLabel());
}
}
```

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Updated Price after 15.0% discount: Rs.127500.0

Product 1: 0 - Unknown - Rs.0.0

Product 2: 101 - Laptop - Rs.127500.0

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Q4

```
public class Employee {  
  
    int empId;  
    String empName;  
    String department;  
    double basicSalary;  
  
    Employee(int id, String name, String dept, double salary) {  
        empId = id;  
        empName = name;  
        department = dept;  
        basicSalary = salary;  
    }  
  
    double calculateNetSalary(double bonusPercentage) {  
        double netSalary;  
  
        netSalary = basicSalary + (basicSalary * bonusPercentage / 100);  
  
        return netSalary;  
    }  
  
    String getEmployeeSummary() {  
        return "Employee ID: " + empId +  
            ", Name: " + empName +  
            ", Department: " + department;  
    }  
  
    void promote(String newDepartment) {  
        department = newDepartment;  
  
        System.out.println(empName +  
            " has been promoted to " +  
            newDepartment + ".");  
    }  
  
    Run | Debug  
    public static void main(String[] args) {  
  
        Employee emp1 = new Employee(id: 101, name: "Kasun", dept: "IT", salary: 80000);  
        Employee emp2 = new Employee(id: 102, name: "Nimal", dept: "HR", salary: 65000);  
  
        System.out.println(emp1.getEmployeeSummary());  
        System.out.println(emp2.getEmployeeSummary());  
  
        System.out.println("Net Salary of " + emp1.empName +  
            ": Rs." + emp1.calculateNetSalary(bonusPercentage: 20));  
  
        System.out.println("Net Salary of " + emp2.empName +  
            ": Rs." + emp2.calculateNetSalary(bonusPercentage: 20));  
  
        emp1.promote(newDepartment: "Management");  
    }  
}
```

```
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Employee ID: 101, Name: Kasun, Department: IT  
Employee ID: 102, Name: Nimal, Department: HR  
Net Salary of Kasun: Rs.96000.0  
Net Salary of Nimal: Rs.78000.0  
Kasun has been promoted to Management.
```

Q5

```
public class BankAccount {  
    int accountNo;  
    String holderName;  
    double balance;  
  
    public BankAccount() {  
        accountNo = 1001;  
        holderName = "Unknown";  
        balance = 0.00;  
    }  
  
    public BankAccount(int accountNo, String holderName, double balance) {  
        this.accountNo = accountNo;  
        this.holderName = holderName;  
        this.balance = balance;  
    }  
  
    public void deposit(double amount) {  
        balance = balance + amount;  
        System.out.println("Updated balance: " + balance);  
    }  
  
    public String withdraw(double amount) {  
        if (balance >= amount) {  
            balance = balance - amount;  
            return "Withdrawal successful. Remaining balance: " + balance;  
        } else {  
            return "Insufficient funds.";  
        }  
    }  
  
    public double getBalance() {  
        return balance;  
    }  
  
    public String getAccountInfo(String branch) {  
        return "Account No: " + accountNo +  
            ", Holder Name: " + holderName +  
            ", Branch: " + branch;  
    }  
  
    Run | Debug  
    public static void main(String[] args) {  
        BankAccount account1 = new BankAccount();  
  
        BankAccount account2 = new BankAccount(accountNo: 2002, holderName: "Kamal Perera", balance: 50000.00);  
  
        System.out.println("Account 1 Information:");  
        System.out.println(account1.getAccountInfo(branch: "Colombo"));  
        account1.deposit(amount: 10000.00);  
        System.out.println(account1.withdraw(amount: 5000.00));  
        System.out.println("Current Balance: " + account1.getBalance());  
  
        System.out.println();  
  
        System.out.println("Account 2 Information:");  
        System.out.println(account2.getAccountInfo(branch: "Kandy"));  
        account2.deposit(amount: 15000.00);  
        System.out.println(account2.withdraw(amount: 20000.00));  
        System.out.println("Current Balance: " + account2.getBalance());  
    }  
}
```

```
Account No: 1001, Holder Name: Unknown, Branch: Colombo  
Updated balance: 10000.0  
Withdrawal successful. Remaining balance: 5000.0  
Current Balance: 5000.0
```

```
Account 2 Information:  
Account No: 2002, Holder Name: Kamal Perera, Branch: Kandy  
Updated balance: 65000.0  
Withdrawal successful. Remaining balance: 45000.0  
Current Balance: 45000.0
```